



Unified Audio-Visual Modeling for Recognizing Which Face Spoke When and What in Multi-Talker Overlapped Speech and Video

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Abstract

We have developed a model that jointly recognizes which face spoke “when” and “what” from multi-talker overlapped speech and video of multiple speakers. For understanding of videos in which multiple speakers are speaking, it is important to recognize which face spoke “when” and “what” from the overlapped speech and multiple speakers’ videos. Conventional methods have to combine speech separation, active speaker detection, and audio-visual speech recognition to address this task. However, the combined system makes the system complex and suboptimal. To address this problem, our idea is to serialize “which face spoke when and what” of multiple speakers into a single token sequence, which is recursively estimated with the proposed unified model using multiple speakers’ videos and overlapped speech as input. Experimental results demonstrate the validity of the proposed method.

Index Terms: overlapped speech recognition, audio-visual speech recognition, active speaker detection

1. Introduction

For a comprehensive understanding of videos in which multiple speakers are speaking, it is important to recognize which face spoke “when” and “what”. Conversations such as meetings and chats typically include speech where multiple people are speaking simultaneously, so the video and speech in such situations include overlapped speech and simultaneously moving multiple faces, where it can be difficult to determine which face is speaking and what is being said.

Several studies have tackled partial problems related to the task of recognizing which face spoke “when” and “what” from overlapped speech and multiple videos. The typical partial problem is to estimate what is spoken from a single video of the speaker’s mouth region, which is known as audio-visual speech recognition (AVSR). In [1–5], transcription of single-talker speech is estimated using the given video of the speaker. For addressing the overlapped speech, a popular approach is to train a model which transcribes the speech of the target speaker specified by the videos. In [6, 7], AVSR of overlapped speech was conducted by using videos of the speakers as keys to implicitly separate overlapping speech.

In practice, multiple videos corresponding to all the people present in the scene are obtained. Thus, we have to decide which face to use before AVSR. The task to estimate which face is speaking is called active speaker detection [8–15]. In [15], to estimate which face spoke “what”, an attention layer is introduced to the encoder, which computes the attention score over multiple face tracks for each audio frame. Although the conventional studies have succeeded in estimating which face is speaking, they only address a single-talker speech, where the

start and end time of the utterance is clear and only a single speaker’s video corresponds to the speech. In practice, the video can include multiple speaking faces as well as non-speaking faces, leading to overlapped speech caused by multiple speakers speaking simultaneously. Thus, to recognize which face spoke “when” and “what” from overlapped speech and multiple videos, we have to combine another system such as speech separation [16–20]. However, the combined system makes the system complex and suboptimal. To the best of our knowledge, no prior research has investigated a unified model that estimates which face spoke “when” and “what” from overlapped speech and videos of multiple speakers.

To address this problem, our idea is to recursively estimate “which face spoke when and what” of multiple speakers utilizing overlapped speech and all videos of multiple speakers as input instead of the conventional approach that transcribes the single-speaker speech specified by a video. For the autoregressive modeling, we serialize transcriptions, timestamp tags [21, 22], and newly introduced video tags of multiple speakers in the same way as serialized output training (SOT) [23]. The video tag is used to associate which face corresponds to the transcription and the timestamp tag represents the start and end time of the utterance.

In this paper, we propose the unified modeling of recognizing which face spoke “when” and “what” in multi-talker overlapped speech and video. The proposed method estimates the serialized token sequence using multiple speakers’ videos and overlapped speech as input. The serialized token sequence consists of multiple speaker’s transcriptions, timestamp tags, and video tags that associate each transcription to the corresponding video. In our experiments, we introduce two new objective criteria to evaluate the prediction performance of which face spoke “when”, and which face spoke “what”. The experimental results demonstrate that the proposed method outperforms the combined system that separately estimates “when and what” is spoken and which face is speaking.

2. Preliminaries

2.1. Multi-talker ASR with SOT

Our proposed method is related to SOT, which was originally proposed to address overlapped speech in audio-only ASR [23]. In this section, we provide a brief explanation of SOT. We denote the acoustic feature of the input speech as $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_T)$, where $\mathbf{x}_t \in \mathbb{R}^F$ denotes the t th frame of the feature, F denotes its dimension, and T denotes the length of acoustic features. We denote utterance-level textual tokens of multiple speakers as $\mathbf{W}^{1:K} = (\mathbf{W}^1, \dots, \mathbf{W}^K)$, where K denotes the number of speakers in the overlapped speech, $\mathbf{W}^k = (w_1^k, \dots, w_{N^k}^k)$ denotes the k th speaker’s textual to-

kens, N^k denotes the length of the token, $w_n^k \in \mathcal{V}$ denotes the n th textual token of the k th speaker, and $\mathcal{V}$ denotes the vocabulary set. SOT was originally proposed for multi-talker ASR without using speech separation, where the model recursively transcribes each utterance of multiple speakers with a single output layer. SOT enables this by introducing a special token [sep] representing speaker change and transcribing the utterance of each speaker in first-in, first-out order [23, 24], where the transcription is estimated in the order of their utterance start time. To recognize multiple utterances with a single output layer, $\mathbf{W}^{1:K}$ is serialized into a single token sequence with special symbol [sep] representing speaker change. The serialized token $\mathbf{S} \in \{\mathcal{V} \cup \mathcal{O}\}$ is given as

$$\mathbf{S} = (w_1^1, \dots, w_{N^1}^1, [\text{sep}], w_1^2, \dots, w_{N^2}^2, [\text{sep}], \dots, w_{N^{K-1}}^{K-1}, [\text{sep}], w_1^K, \dots, w_{N^K}^K, [\text{eos}]), \quad (1)$$

where [eos] denotes the end of a sentence, $\mathcal{O} = \{[\text{sep}], [\text{eos}]\}$, and we assume that $\mathbf{W}^{1:K}$ is sorted in order of utterance start times for simplicity.

2.2. Joint modeling of multi-talker ASR and timestamp prediction

In [22], SOT was further extended to predict the timestamps of each utterance. We denote the start time token and the end time token of multiple speakers as $\mathbf{T}_s^{1:K} = ([t_s^1], \dots, [t_s^K])$ and $\mathbf{T}_e^{1:K} = ([t_e^1], \dots, [t_e^K])$, respectively, where $[t_s^k] \in \mathcal{T}$ and $[t_e^k] \in \mathcal{T}$ denote the k th utterance's start time token and end time token, respectively, and $\mathcal{T}$ denotes the quantized time token label set. The quantized time token is obtained by rounding the continuous timestamp values to the nearest quantized value every Q seconds. The serialized token $\tilde{\mathbf{S}} \in \{\mathcal{V} \cup \mathcal{T} \cup \mathcal{O}\}$ is given as

$$\tilde{\mathbf{S}} = ([t_s^1], [t_e^1], w_1^1, \dots, w_{N^1}^1, [\text{sep}], [t_s^2], [t_e^2], w_1^2, \dots, w_{N^2}^2, [\text{sep}], \dots, [t_s^K], [t_e^K], w_1^K, \dots, w_{N^K}^K, [\text{eos}]). \quad (2)$$

Multi-talker ASR with SOT estimates the generation probability of $\tilde{\mathbf{S}}$ given $\mathbf{X}$ as follows:

$$P(\tilde{\mathbf{S}}|\mathbf{X}; \Theta_{\text{MT}}) = \prod_{l=1}^{|\tilde{\mathbf{S}}|} P(\tilde{s}_l | \tilde{s}_{1:l-1}, \mathbf{X}; \Theta_{\text{MT}}), \quad (3)$$

where $\tilde{s}_l$ denotes the l th token of $\tilde{\mathbf{S}}$, $\tilde{s}_{1:l-1} = (\tilde{s}_1, \dots, \tilde{s}_{l-1})$, $|\tilde{\mathbf{S}}|$ denotes the length of $\tilde{\mathbf{S}}$, and Θ_{MT} denotes the parameter of the multi-talker ASR model. The parameter Θ_{MT} is optimized with the following cross-entropy function:

$$L_{\text{MT}} = -\log P(\tilde{\mathbf{S}}|\mathbf{X}; \Theta_{\text{MT}}). \quad (4)$$

3. Proposed Method

3.1. Strategy

As discussed in Section 1, for a comprehensive understanding of situations where multiple speakers are speaking, addressing the overlapped speech and multiple videos with a unified model is imperative. To address the overlapped speech and multiple speakers' videos to estimate which face spoke "when" and "what", we train the model so that it estimates the serialized token sequence that reveals "which face spoke when and what" of multiple speakers utilizing overlapped speech and all the videos of multiple speakers.

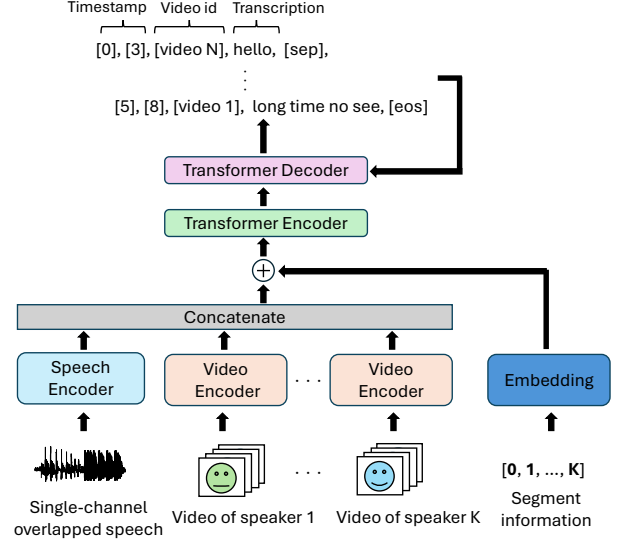


Figure 1: Overview of proposed method.

Figure 1 shows an overview of the proposed method. The serialized token sequence consists of transcriptions, timestamp tags, and video tags that associate transcriptions to the corresponding video for each utterance. The model is trained to estimate the serialized token sequence recursively with a single output layer using SOT. We train the model using cross-entropy loss w.r.t. the token sequence.

3.2. Formulation

We denote the video tag that indicates the corresponding video frames for each transcription as $\mathbf{u} = u_1, \dots, u_K$, where $u_k \in \mathcal{U}$ denotes the video tag for the k th utterance. For autoregressive modeling of transcriptions, timestamp tags, and video tags, we serialize them into a single sequence. When the video tag is placed in front of each transcription, the serialized sequence $\tilde{\mathbf{S}} \in \{\mathcal{V} \cup \mathcal{T} \cup \mathcal{O} \cup \mathcal{U}\}$ is obtained as

$$\tilde{\mathbf{S}} = ([t_s^1], [t_e^1], u^1, w_1^1, \dots, w_{N^1}^1, [\text{sep}], [t_s^2], [t_e^2], u^2, w_1^2, \dots, w_{N^2}^2, [\text{sep}], \dots, [t_s^K], [t_e^K], u^K, w_1^K, \dots, w_{N^K}^K, [\text{eos}]). \quad (5)$$

Note that video tags can be placed after transcriptions instead of before, which can affect prediction performance for the autoregressive modeling. We discuss the effect of this ordering in Section 4.4.

3.3. Modeling

We use a Transformer-based model. We denote the video frames of multiple speakers as $\mathbf{V}^{1:K} = (\mathbf{V}^1, \dots, \mathbf{V}^K)$, where $\mathbf{V}^k = (\mathbf{V}_1^k, \dots, \mathbf{V}_M^k)$ denote the k th speaker's video frames, $\mathbf{V}_m^k \in \mathbb{R}^{W \times H \times C}$ denotes the m th frame of each video, M denotes the video length, and W , H , and C denote the width, height, and channel of each frame, respectively. Since we input all the video frames corresponding to the overlapped speech, M is identical among all videos. To explicitly indicate which parts correspond to speech and which parts correspond to the videos of each speaker, we use the segment information $\mathbf{g} = (\mathbf{0}, \mathbf{1}, \dots, \mathbf{K})$, where $\mathbf{0} = (0, \dots, 0)$ denotes the speech part whose length is T and $\mathbf{k} = (k, \dots, k)$ denotes the video part of the k th speaker whose length is M . The joint generation

probability is obtained as follows:

$$\mathbf{H}_s = \text{SpeechEnc}(\mathbf{X}; \boldsymbol{\theta}_{\text{senc}}), \quad (6)$$

$$\mathbf{H}_v^k = \text{VideoEnc}(\mathbf{V}^k; \boldsymbol{\theta}_{\text{venc}}), \quad (7)$$

$$\mathbf{E} = \text{Embed}(\mathbf{g}; \boldsymbol{\theta}_{\text{emb}}), \quad (8)$$

$$\mathbf{H} = \text{Concat}(\mathbf{H}_s, \mathbf{H}_v^1, \dots, \mathbf{H}_v^K) + \mathbf{E}, \quad (9)$$

$$\mathbf{G} = \text{TransformerEnc}(\mathbf{H}; \boldsymbol{\theta}_{\text{enc}}), \quad (10)$$

$$P(\tilde{s}_l | \tilde{s}_{1:l-1}; \boldsymbol{\Theta}) = \text{TransformerDec}(\mathbf{G}, \tilde{s}_{1:l-1}; \boldsymbol{\theta}_{\text{dec}}), \quad (11)$$

where $\text{SpeechEnc}(\cdot)$ is a speech encoder, $\boldsymbol{\theta}_{\text{senc}}$ denotes its parameter, $\text{VideoEnc}(\cdot)$ is a video encoder, $\boldsymbol{\theta}_{\text{venc}}$ denotes its parameter, $\text{Embed}(\cdot)$ is an embedding layer, $\boldsymbol{\theta}_{\text{emb}}$ denotes its parameter, $\text{Concat}(\cdot)$ denotes the concatenation along the time axis, $\text{TransformerEnc}(\cdot)$ is a Transformer encoder, $\boldsymbol{\theta}_{\text{enc}}$ denotes its parameters, $\text{TransformerDec}(\cdot)$ is a Transformer decoder, $\boldsymbol{\theta}_{\text{dec}}$ denotes its parameters, $\text{Linear}(\cdot)$ denotes a linear layer with softmax activation, $\boldsymbol{\theta}_{\text{linear}}$ denote its parameter, $\tilde{s}_l$ denotes the l th token of $\tilde{\mathbf{S}}$, and $\tilde{s}_{1:l-1} = (\tilde{s}_1, \dots, \tilde{s}_{l-1})$. For time alignment, we utilize positional encoding to the output of the speech encoder and video encoder. The detailed architecture of the model is described in Section 4.2. The generation probability of $\tilde{\mathbf{S}}$ is obtained as Eq. (3). The parameter $\boldsymbol{\Theta} = \{\boldsymbol{\theta}_{\text{senc}}, \boldsymbol{\theta}_{\text{venc}}, \boldsymbol{\theta}_{\text{emb}}, \boldsymbol{\theta}_{\text{enc}}, \boldsymbol{\theta}_{\text{dec}}\}$ is optimized with the following cross-entropy function:

$$L = -\log P(\tilde{\mathbf{S}} | \mathbf{X}, \mathbf{V}^1, \dots, \mathbf{V}^K; \boldsymbol{\Theta}). \quad (12)$$

4. Experiment

4.1. Dataset

We used the Lip Reading Sentences 3 (LRS3) dataset [25] for our experiments. Although the sentences of LRS3 in the train-val and test sets are clipped to 100 characters or 6 seconds, the pre-train set includes longer videos, which leads to a gap between the training set and the others. Thus, we mixed the original pre-train, train-val, and test sets and split them into new pre-train, train-val, and test sets so that speakers do not overlap between datasets and the train-val and test sets include longer audio. Since LRS3 consists of single-talker videos, we created two-speaker and three-speaker simulated mixtures by mixing the utterances of different speakers. When mixing the audio signals, the original volume of each utterance was kept unchanged, resulting in an average signal-to-interference ratio of about 0 dB. The delay values applied to each utterance were randomly chosen under the constraints presented in [23]. The start times of individual utterances differed by 0.5 s or longer so that every utterance in each mixed audio sample had at least one speaker-overlapped region with other utterances. Since the length of the created overlapped speech exceeds that of each speaker’s video, we copied the nearest frame of the video to fill the empty frames. For the memory constraint, we only used one frame every five frames, which resulted in 5 fps videos. We clipped 96×96 regions around mouth and converted them to gray scale. We added Gaussian noise of standard deviation 0.05 to every frame of the video.

We used 80 log mel-scale filterbank coefficients as acoustic features, which were extracted using a 20-ms-long Hann window with a 10-ms-long shift. We used 8K sentence pieces as textual tokens. We set Q to 0.5, which means continuous timestamps were rounded every 0.5 s.

4.2. Implementation

For the video encoder, we used MobileNetV3-Small [26]. The speech encoder consists as follows. First, the acoustic feature was passed to layers composed of two 1×1 convolutions with 1×1 strides, two max pooling with a stride of 2, two 3×3 depthwise convolutions with 1×1 strides, and two long-short term memory layers with outputs of 256 dimensions. Next, we stacked eight Transformer encoder blocks, where the number of heads in the multi-head attention was set to 4, the dimensions of the output continuous representations were set to 256, and the dimensions of the inner output in the position-wise feed-forward networks were set to 1,024. Before concatenation in Eq. (9), we added positional encoding to the output of the speech encoder and the video encoder so that the model captures the temporal alignment. The dimension of the embedding in Eq. (8) was 256. We used a Transformer-based model [27,28] in this paper. For the Transformer encoder in Eq. (10) and the Transformer decoder in Eq. (11), we stacked two-layer Transformer encoder blocks and two-layer Transformer decoder blocks, respectively, where the settings were the same as those of the speech encoder. For the activation function, we used Swish [29].

4.3. Settings

We compared the following five methods: conventional single-talker ASR, conventional single-talker AVSR, combined system of multi-talker ASR and lip moving detection, combined system of multi-talker AVSR and lip moving detection, and the proposed method. The single-talker models were trained with single speaker speech and the corresponding video while the multi-talker models were trained with all the speech and the corresponding videos including single speaker, two speakers, and three speakers. For the fair comparison, the architecture of all the models was the same. The multi-talker AVSR is a model that adds multiple video inputs to the multi-talker ASR model, which estimates transcriptions and timestamp tags without video tags from overlapped speech and videos of multiple speakers. As a naive approach to estimate which face is speaking using the multi-talker ASR or multi-talker AVSR, we experimented with a combined system utilizing a lip moving detection model. The lip moving detection model, which consists of the video encoder described in Section 4.2, four Transformer-encoder blocks, and a classification layer, estimates “1” when the lip is moving (otherwise, zero) in a frame-wise manner. It was trained with the multi-talker training data described in Section 4.1 so that the copied frames were predicted as zero and its accuracy on test data was 97.6%. To estimate video tags with the lip moving detection model, we calculated the match rate of all possible pairs of the predicted lip movement of each video and the predicted timestamp tag of each utterance, and assigned the video tag to each utterance with the highest match rate allowing duplication.

To investigate the effect of the ordering of the transcriptions and video tags of the proposed method, we compared the following three methods: the proposed method without transcription, the proposed method with video tags in front of each transcription (Proposed method w/ tag front), and the proposed method with video tags after each transcription (Proposed method w/ tag end). In the proposed method without transcription, the model was trained to recursively estimate the quantized timestamp tokens $\mathbf{T}_s^{1:K}$ and $\mathbf{T}_e^{1:K}$, the speaker change token [sep], and the video tags $\mathbf{u}$.

The video encoder was pretrained with a lip reading task

Table 1: *Evaluation results.*

Number of speakers in test dataset	Methods	WER (%)	VTER (%)	VWER (%)
1	Single-talker ASR	19.8	1.3	19.8
	Single-talker AVSR	18.8	1.1	18.8
	Multi-talker ASR + lip moving detection	19.5	1.2	19.5
	Multi-talker AVSR + lip moving detection	18.3	1.2	18.3
	Proposed method (w/o transcription)	–	1.1	–
	Proposed method (w/ tag front)	18.3	1.1	18.3
2	Proposed method (w/ tag end)	18.7	1.1	18.7
	Multi-talker ASR + lip moving detection	27.5	3.0	29.3
	Multi-talker AVSR + lip moving detection	25.8	2.8	27.5
	Proposed method (w/o transcription)	–	1.3	–
	Proposed method (w/ tag front)	26.0	1.5	26.7
3	Proposed method (w/ tag end)	25.9	1.8	26.4
	Multi-talker ASR + lip moving detection	35.6	6.4	39.7
	Multi-talker AVSR + lip moving detection	34.1	5.6	37.0
	Proposed method (w/o transcription)	–	1.6	–
	Proposed method (w/ tag front)	34.4	2.6	35.6
	Proposed method (w/ tag end)	34.2	2.5	35.6

using the single-talker speech and video of the LRS3 pre-train dataset. The output of the video encoder was fed into the two-layer Transformer decoder blocks to transcribe “what” is said only from video of the speaker. The speech encoder was pre-trained with a multi-talker ASR task utilizing the multi-talker speech training data of the LRS3 pre-train dataset in Section 4.1. The output of the speech encoder was fed into the two-layer Transformer decoder blocks to transcribe “what” is said only from speech.

All models were optimized using the RAdam [30] algorithm. The speech encoder and lip moving detection model was trained with a mini-batch size of 32 and the learning rate of $1e-4$. The video encoder was trained with a mini-batch size of 16 and the learning rate of $1e-4$. All other models were trained with a mini-batch size of 8 and the learning rate of $1e-5$ for the memory constraint. All parameters including that of the speech encoder and video encoder were trainable. The training steps were stopped if the loss on the validation set did not decrease for ten epochs in succession.

We used the word error rate (WER), and the newly introduced visual word error rate (VWER) and visual time error rate (VTER) to evaluate the overall performance of the methods. In contrast to the original WER that only considers “what” is spoken, VWER is a criterion that evaluates the prediction performance of which face spoke what. In VWER, every video has a label transcription, and the predicted video tag’s transcription become the label for evaluating WER. In other words, when the predicted video tag is not correct, the predicted transcriptions are counted as error even when they match the transcriptions of another video. VTER is a criterion that evaluates the prediction performance of which face spoke when. In VTER, every video has an utterance-level timestamp label and the predicted video tag’s timestamp label become the label to calculate TER. TER was calculated as the sum of missed speaker rate and false alarm rate [22]. When comparing hypothesized boundaries to references, we used a tolerance of ± 250 ms. When calculating the original WER in multi-talker overlapped ASR settings, we compared hypotheses with references while considering the order of utterances. We only evaluated textual tokens excluding the special token $\mathcal{O}$, the timestamp token $\mathcal{T}$, and the video tag token $\mathcal{U}$.

4.4. Results

Table 1 shows the result. Note that WER is relatively high compared to other studies using LRS3 because we used a different test set that includes longer audio, as explained in Section 4.1. We can see that the proposed method outperforms the combined system in terms of both VTER and VWER. Since the difference between VWER and WER arises from the prediction error of video tags, the combined system makes more errors on video tag assignment compared to the proposed method. We assume this is because although the combined system that utilizes the independently predicted lip movement and timestamp tags accumulate errors, the proposed method is directly optimized to estimate it.

We can see that the proposed method (w/ tag front) and the proposed method (w/ tag end) achieve almost the same performance in terms of both VTER and VWER while the proposed method (w/o transcription) outperforms them in terms of VTER. We assume this is because the prediction error of transcription causes the model to assign incorrect video tags. However, transcriptions are expected to help estimating video tags in more challenging scenes where the speakers of some videos do not speak and the speakers not included in the video speak, which remains our future work.

5. Conclusion

In this paper, we proposed a unified model that jointly recognizes which face spoke “when” and “what” from multi-talker overlapped speech and video of multiple speakers. Most conventional methods have transcribed “what” is spoken from the given video frames, but in videos where multiple speakers are simultaneously speaking, figuring out which face spoke “when” and “what” is practically important. To address this problem, we proposed the unified model that is trained to estimate the serialized token sequence revealing which face spoke “when” and “what” from videos of multiple speakers and overlapped speech. To evaluate the overall performance, we used newly introduced objective criteria named VWER and VTER in addition to WER. Experimental results showed that although the proposed method is a single model that is much simpler than the combined system, it delivered the superior performance.

6. References

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